

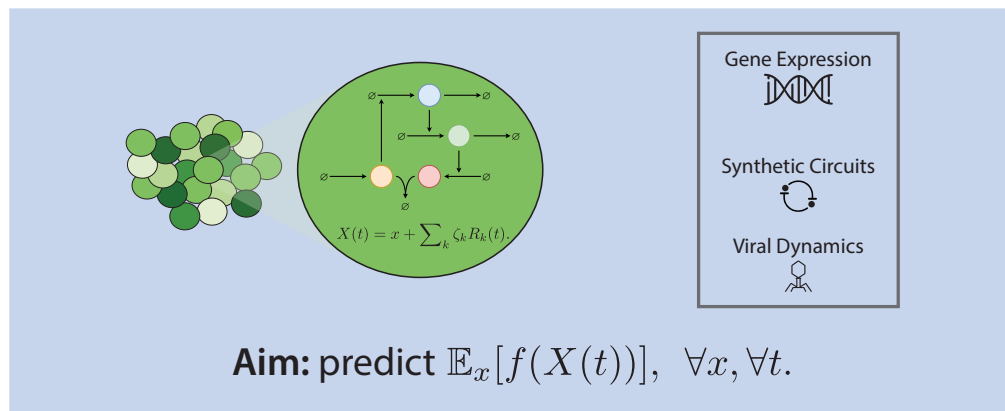
Interpretable neural approximation of stochastic reaction dynamics with guaranteed reliability

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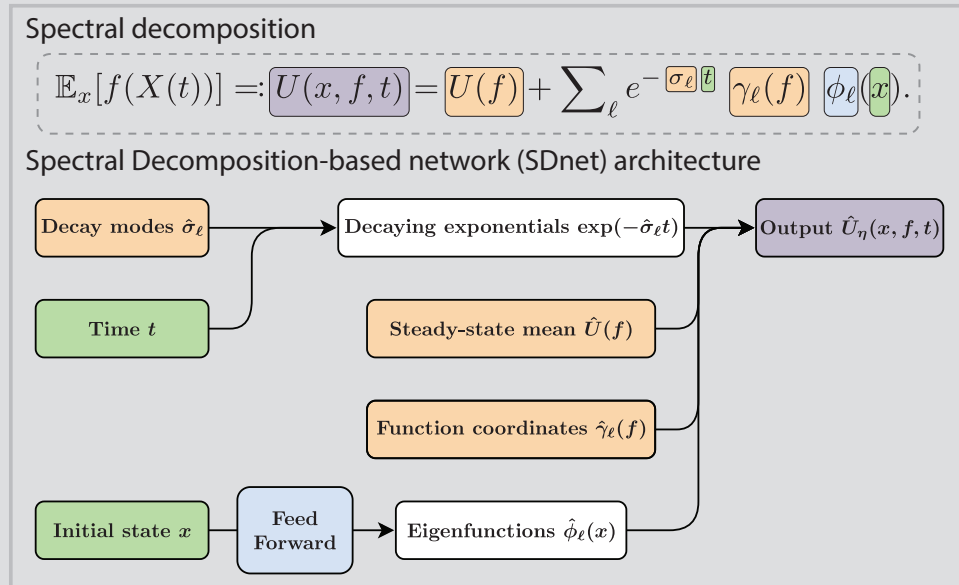
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1 Introduction



2 Methods

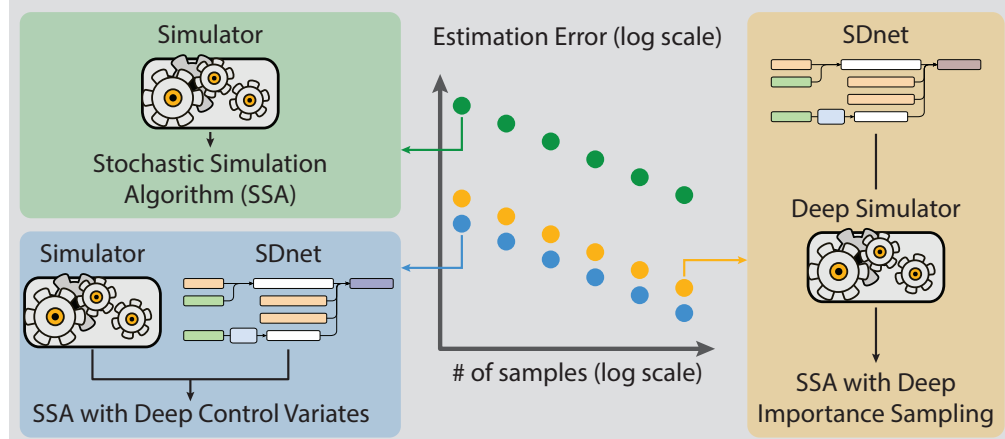
Interpretable SDnet architecture



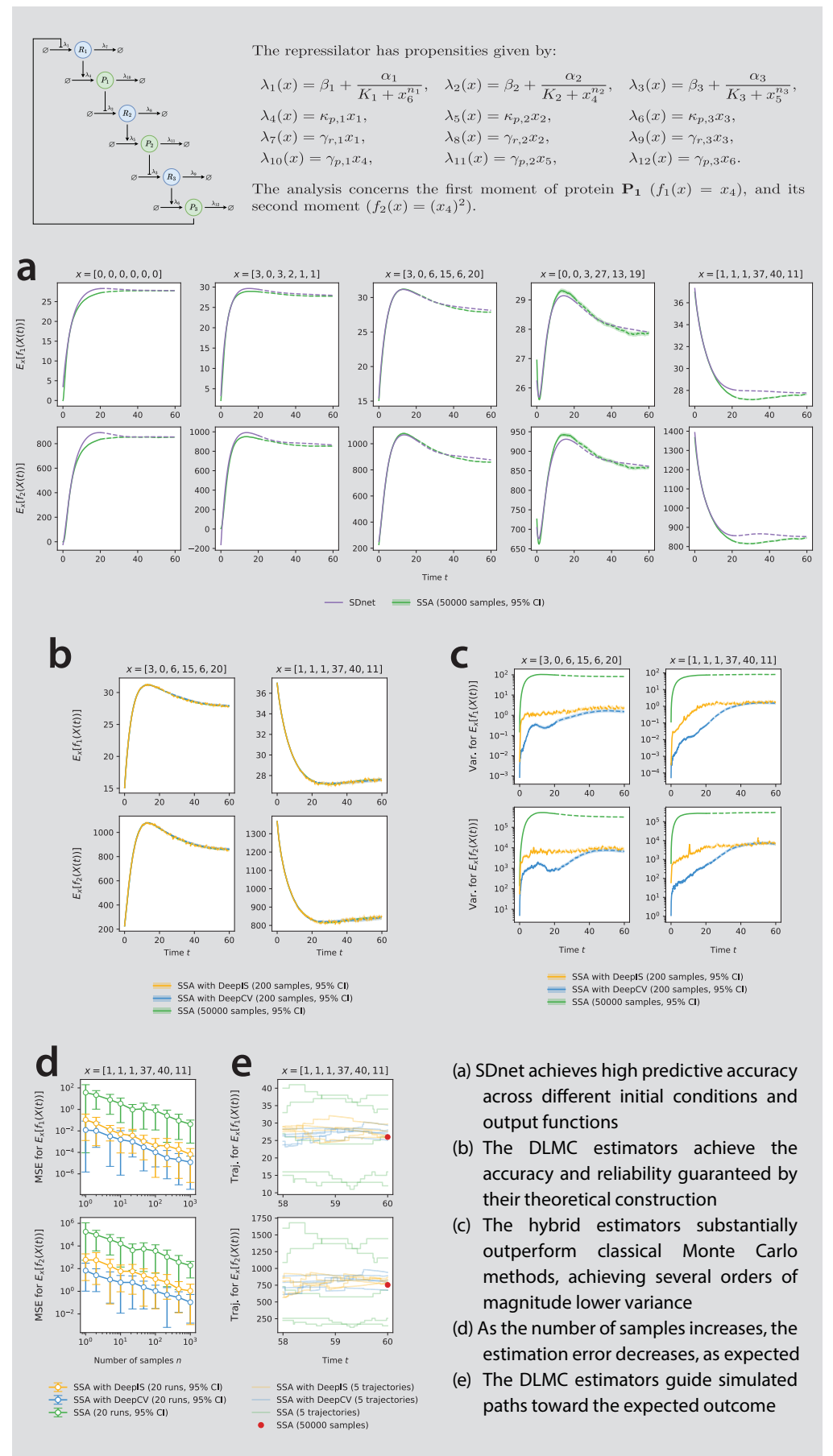
$$\text{Loss} = \text{Distance} \left(f(X(t_{\beta})), \hat{U}_{\eta}(X(t_{\alpha}), f, t_{\beta} - t_{\alpha}) + \int_{t_{\alpha}}^{t_{\beta}} \Delta \hat{U}_{\eta}(X(s), f, t_{\beta} - s) d\tilde{R}(s) \right).$$



Reliable Deep Learning/Monte Carlo estimation



3 Results



References

- Gupta, A., Schwab, C. & Khammash, M. DeepCME: A deep learning framework for computing solution statistics of the chemical master equation. PLoS Computational Biology 17, e1009623 (2021).
- Gupta, A. & Khammash, M. Sparse spectral estimation of the Koopman operator for stochastic reaction networks. Manuscript in preparation.